# UAT Rules Inventory — IAM Automation Phase 1

## Approved product boundary

Phase 1 is a **read-only identity reconciliation and audit platform**. It ingests approved exports, normalizes records, matches identities, builds an internal canonical analytical employee/worker representation, classifies findings, preserves evidence/history, supports human review, and produces reports/evidence exports.

Phase 1 may mutate internal analytical/runtime state required for ingestion, reconciliation, lineage, review state, evidence, local notifications, backups, and technical snapshot distribution. It must not make authoritative IAM decisions or mutate authoritative source platforms. Humans decide; authoritative platforms execute.

Scope classifications:

- **PHASE1_CORE** — required to prove reconciliation, finding classification, evidence, review, and exports.
- **PHASE1_PLATFORM** — technical capability required to operate Phase 1 safely.
- **PHASE2_WRITER** — performs, approves, applies, or orchestrates authoritative business changes.
- **FUTURE_INTEGRATION** — future connector or downstream capability, not required for Phase 1 acceptance.
- **OUT_OF_SCOPE** — repository behavior not part of current acceptance.

Markers:

- **BUSINESS DECISION REQUIRED** — expected finding interpretation or policy is not approved.
- **ASSUMPTION CURRENTLY ENCODED** — code selects an outcome without clear business approval.
- **CONTRADICTION** — repository sources disagree; this does not expand Phase 1 scope.

## Classification summary

| Scope | Count | Phase 1 sign-off treatment |
|---|---|---|
| PHASE1_CORE | 34 | In scope; may be blocked by business discovery or technical defects. |
| PHASE1_PLATFORM | 8 | In scope as safety/operational acceptance, not as a decision engine. |
| PHASE2_WRITER | 5 | Deferred; must not block Phase 1 sign-off. |
| FUTURE_INTEGRATION | 4 | Deferred; track separately from Phase 1 acceptance. |
| OUT_OF_SCOPE | 1 | Do not include in current UAT. |
| **Total classified rules** | **52** | |

## Rules inventory

| ID | Scope | Business area | Rule / requirement | Source file(s) / function / module | Evidence and current status | IAM validation | Risk if misunderstood |
|---|---|---|---|---|---|---|---|
| R-001 | PHASE1_CORE | Product scope | Phase 1 detects and explains source observations; humans decide and source platforms execute. | `ARCHITECTURE.md`; `README.md`; application routes | Approved architecture clarification; replaces earlier mixed-scope interpretation. | No — scope approved; verify technical enforcement. | Writer behavior may be mistaken for product acceptance. |
| R-002 | PHASE1_CORE | Ingestion | Master Roster staging requires Employee Number and Employee Name. | `application/master_roster_ingestion_service.py`; `ingest_master_roster` | Implemented/tested; source-specific required fields still need confirmation. | Yes | Invalid rows may enter reconciliation or be silently excluded. |
| R-003 | PHASE1_CORE | Ingestion | Roster Lock rows are staged using vendor name and vendor employee ID. | `application/roster_lock_ingestion_service.py`; `ingest_roster_lock` | Implemented; **ASSUMPTION CURRENTLY ENCODED** that the composite key is correct. | Yes | Rows can be lost or treated as new identities. |
| R-004 | PHASE1_CORE | Ingestion | AccessNow CSV/XLSX rows are grouped by AccessNow employee ID. | `application/accessnow_ingestion_service.py`; AccessNow ingestion | Implemented/tested. | Yes | Multiple observations may be collapsed incorrectly. |
| R-005 | PHASE1_CORE | Ingestion | RSR requires supplier reference, worker ID, and worker fields. | `application/rsr_supplier_worker_details.py`; RSR staging | Implemented; **ASSUMPTION CURRENTLY ENCODED** until source owner confirms required fields. | Yes | Valid supplier evidence may be quarantined. |
| R-006 | PHASE1_CORE | Identity matching | Strong matching signals include AccessNow employee ID, RSR worker ID, vendor employee ID, buyer reference, and Master identifiers. | `application/source_reconciliation.py`; `_match_types` | Implemented; equivalence is not fully approved — **ASSUMPTION CURRENTLY ENCODED**. | Yes | Wrong observations may be joined or split. |
| R-007 | PHASE1_CORE | Identity matching | NUID, employee number, vendor ID, and buyer reference are aliased in matching. | `source_reconciliation.py`; `_match_types` | Implemented; **ASSUMPTION CURRENTLY ENCODED** that aliases mean identity equivalence. | Yes | Unrelated people may be treated as one analytical identity. |
| R-008 | PHASE1_CORE | Identity matching | Multiple strong matches surface a conflict rather than silently selecting one. | `source_reconciliation.py`; RSR matching tests | Implemented/tested. | Yes | Ambiguity may become a false positive canonical match. |
| R-009 | PHASE1_CORE | Identity matching | Email and exact normalized first/last name are fallback signals. | `source_reconciliation.py`; fallback matching | Implemented; **ASSUMPTION CURRENTLY ENCODED**. | Yes | Shared email or common names can create false matches. |
| R-010 | PHASE1_CORE | Identity matching | No fuzzy score or confidence threshold is currently available. | `source_reconciliation.py` | Confirmed implementation gap; possible matches must be surfaced without unsafe merge. | Yes | Possible duplicates may be inconsistently triaged. |
| R-011 | PHASE1_CORE | Canonical analytical identity | Canonical Worker/Employee is an internal analytical identity used to reconcile observations, not an HR, Fieldglass, AccessNow, or provisioning record. | `docs/architecture/WORKER_EMPLOYEE_CANONICAL_ARCHITECTURE.md`; `orm.py` | Approved terminology clarification; Worker, identifiers, engagements, profiles, and lineage are analytical structures. | No — verify wording in outputs. | Internal identity may be mistaken for authoritative employment or access status. |

| ID | Scope | Business area | Rule / requirement | Source file(s) / function / module | Evidence and current status | IAM validation | Risk if misunderstood |
|---|---|---|---|---|---|---|---|
| R-012 | PHASE2_WRITER | Canonical promotion | Authoritative promotion of an unmatched person into an operational employee is writer behavior, not Phase 1. | `rsr_supplier_worker_details.py`; `employee_service.py:_promote_record` | Existing code can create an ACTIVE Worker; classify **DEFERRED — PHASE2_WRITER**. Phase 1 may create/resolve an analytical identity only under approved matching evidence. | No for Phase 1; writer owner later | UAT could approve a business outcome that Phase 1 must not make. |
| R-013 | PHASE1_CORE | Duplicate detection | Canonical duplicate candidates are surfaced using normalized identity signals, including full name. | `employee_service.py: list_canonical_duplicates` | Implemented; name grouping is a candidate signal, not proof — **ASSUMPTION CURRENTLY ENCODED**. | Yes | Same-name people may be incorrectly presented as one identity. |
| R-014 | PHASE2_WRITER | Merge | Authoritative merge execution is a future writer action; evidence-preserving merge design is retained for later phase. | `employee_service.py: merge_workers`; merge docs/tests | Existing route/service preserves evidence but is not Phase 1 acceptance — **DEFERRED — PHASE2_WRITER**. | No for Phase 1 | False merge can corrupt authoritative identity or downstream access. |
| R-015 | PHASE2_WRITER | Merge governance | Automatic or authoritative merge requires a future approval/reversibility policy. | merge plan docs; `employee_routes.py` | Existing review/merge path does not become a Phase 1 requirement — **DEFERRED — PHASE2_WRITER**. | No for Phase 1 | Phase 1 users may be treated as authorized identity decision-makers. |
| R-016 | PHASE1_CORE | Finding classification | A field conflict does not automatically require a winner; it may be authority, discrepancy, expected difference, source latency, insufficient evidence, or policy-relevant finding. | `report_service.py`; reconciliation services; requirements gaps | New approved analytical model; field-level authority remains **BUSINESS DECISION REQUIRED** where applicable. | Yes | False alerts or silent overwrites. |
| R-017 | PHASE1_CORE | Finding classification | Required Phase 1 classifications are `CONSISTENT`, `EXPECTED_DIFFERENCE`, `SOURCE_LATENCY`, `INSUFFICIENT_EVIDENCE`, `DISCREPANCY`, and `POLICY_RELEVANT_FINDING`. | proposed UAT model; current discrepancy/report structures | Classification vocabulary must be confirmed and mapped to existing statuses — **BUSINESS DECISION REQUIRED**. | Yes | Findings cannot be prioritized or reviewed consistently. |
| R-018 | PHASE1_CORE | Source precedence | A field-level authoritative source may be used only where IAM has approved that field/source rule. | `REQUIREMENTS_GAP_REVIEW.md`; meeting notes; `report_service.py` | No complete matrix found — **BUSINESS DECISION REQUIRED**. | Yes | Analytical view may conceal a real discrepancy. |
| R-019 | PHASE1_CORE | Duplicate handling | Exact same-source observations are summarized; current/superseded treatment must preserve all evidence. | `employee_service.py`; duplicate tests | Implemented using normalized signals and created time; winner semantics remain **BUSINESS DECISION REQUIRED**. | Yes | Late corrections or duplicate rows may be misrepresented. |

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|---|---|---|---|---|---|---|---|
| R-020 | PHASE1_CORE | Duplicate handling | Cross-source corroboration is evidence, not automatic proof of identity. | `employee_service.py` duplicate summary | Current implementation can classify corroboration; **ASSUMPTION CURRENTLY ENCODED** if treated as identity proof. | Yes | False confidence in identity matching. |
| R-021 | PHASE1_CORE | Status observation | Status aliases normalize TERM/TERMINATED/TERMINATION and DEACT/DISABLED/INACTIVE. | `application/status_catalog.py` | Implemented/tested; normalization is analytical, not an authoritative lifecycle action. | Yes | Different source vocabulary may be misread. |
| R-022 | PHASE1_CORE | Status finding | ACTIVE/DEACTIVATED/TERMINATED ranking is an observation heuristic and must not make a lifecycle decision. | `status_catalog.py`; AccessNow ingestion | Code ranks statuses; **ASSUMPTION CURRENTLY ENCODED** until IAM approves interpretation. | Yes | Stale active data may suppress a discrepancy. |
| R-023 | PHASE1_CORE | Status reconciliation | Conflicting status observations must remain explainable with dates/source evidence; AccessNow termination handling is an observation rule. | `accessnow_ingestion_service.py`; status tests | Partial; termination rows can override internal observed status; not an authoritative deactivation. | Yes | A finding may be falsely resolved. |
| R-024 | PHASE1_CORE | Missing/conflicting data | Missing or invalid source rows are rejected/quarantined with recoverable evidence. | ingestion services; quarantine models/tests | Implemented at boundaries; disposition/ownership/SLA are **BUSINESS DECISION REQUIRED**. | Yes | Exceptions disappear or remain unresolved. |
| R-025 | PHASE1_CORE | Unknown observation | An unmatched source record becomes an analytical unmatched finding/candidate; it is not automatically an authoritative employee. | `source_reconciliation.py`; RSR matching | Analytical candidate path exists; existing promotion path is Phase 2 — **CONTRADICTION** with old mixed-scope docs. | Yes | Unknown people may be treated as approved employees. |
| R-026 | PHASE1_CORE | Non-standard population | Phase 1 must detect and report observations outside the standard roster flow; it does not decide eligibility. | requirements gaps; meeting notes; reports | Complete policy absent — **BUSINESS DECISION REQUIRED**. | Yes | Contractors/transfers/exceptions may be invisible or misclassified. |
| R-027 | PHASE1_CORE | Reconciliation coverage | Presence in one source and absence in another is a finding candidate, not automatically an error. | `report_service.py`; coverage fields | Coverage is visible; expected population and absence windows are **BUSINESS DECISION REQUIRED**. | Yes | False positives or missed population gaps. |
| R-028 | PHASE1_CORE | Rehire detection | Phase 1 detects whether evidence meets configured potential-rehire conditions; it does not approve/reject or execute rehire. | `rehire_alert_service.py`; `business_calendar.py` | Alert path exists; policy parameters remain **BUSINESS DECISION REQUIRED**. | Yes | Detection may be mistaken for a lifecycle decision. |

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| R-029 | PHASE1_CORE | Rehire temporal rule | Business/calendar day type, boundary inclusion, holiday calendar, termination date, proposed start date, and fallback timestamps must be explicit inputs to detection. | `rehire_alert_service.py`; `business_calendar.py` | Code currently uses 0–30 business days and created-time fallback — **ASSUMPTION CURRENTLY ENCODED**. | Yes | False or missed rehire findings. |
| R-030 | PHASE1_CORE | Source latency | Refresh cadence, snapshot date, effective date, received/imported date, data age, and acceptable latency must be preserved and distinguishable. | ingestion batch models; source services; meeting notes | Partial; no common latency policy — **BUSINESS DECISION REQUIRED**. | Yes | Stale data can be presented as current. |
| R-031 | PHASE1_CORE | Absence semantics | Absence becomes a discrepancy only after the approved population-specific latency window; before then it may be `SOURCE_LATENCY` or `INSUFFICIENT_EVIDENCE`. | reconciliation/reporting; requirements gaps | No complete window matrix — **BUSINESS DECISION REQUIRED**. | Yes | Excessive alert noise or missed discrepancy. |
| R-032 | PHASE1_CORE | No-hire finding | If an approved no-hire/blacklist export is available, Phase 1 may detect a possible match and present `POLICY_RELEVANT_FINDING`; it must not block, reject, or approve hire. | REQ-003; meeting notes; no complete enforcement service | Detection/enforcement incomplete — **BUSINESS DECISION REQUIRED** for source/match/evidence. | Yes | Ineligible condition may be missed or treated as a decision. |
| R-033 | PHASE1_CORE | Historical identity | Historical identifiers, names, engagements, immutable profile versions, lineage, and source snapshots remain searchable evidence. | `worker_history_service.py`; architecture docs/tests | Implemented/tested. | Yes | Identity changes may erase or confuse evidence. |
| R-034 | PHASE1_CORE | Identifier/name changes | Identifier and name changes are observations to reconcile and version; continuity versus ambiguity must be explainable. | `worker_history_service.py`; `employee_service.py` | History exists; identity continuity policy is **BUSINESS DECISION REQUIRED**. | Yes | Wrong split/merge of analytical identities. |
| R-035 | PHASE1_CORE | Profile/manager changes | Name, manager, role, email, and profile changes must be shown as field-level evidence with timestamps and source, not silently overwritten. | `employee_service.py`; source update preview/tests | Partial selective-update behavior exists; authoritative field rules are **BUSINESS DECISION REQUIRED**. | Yes | Reviewers may miss a material discrepancy. |
| R-036 | PHASE2_WRITER | Lifecycle/access changes | Deactivate, reactivate, role/access change, provision, and deprovision actions in authoritative systems are outside Phase 1. | `change_request.py`; README; requirements | Request/apply paths exist as future code — **DEFERRED — PHASE2_WRITER**. Phase 1 may report conditions only. | No for Phase 1 | UAT could imply access or lifecycle changed. |

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|---|---|---|---|---|---|---|---|
| R-037 | PHASE1_CORE | Export evidence | Reports provide a reproducible analytical view with one row per approved report grain and source coverage/open findings. | `report_service.py`; report tests | Implemented; grain/inclusion rules need stakeholder confirmation. | Yes | Consumers may misread population or row grain. |
| R-038 | FUTURE_INTEGRATION | Salesforce evidence export | Salesforce-format output is a future/manual handoff contract, not a Phase 1 acceptance requirement and never a live provisioning action. | `salesforce_export_service.py`; Salesforce guide | Export-only implementation exists; live API and Salesforce-specific handoff are deferred — **DEFERRED — FUTURE_INTEGRATION**. | No for Phase 1 | Handoff may be mistaken for live account creation. |
| R-039 | PHASE1_CORE | Export completeness | Export completeness means expected analytical population, exclusions, findings, control totals, and reproducibility are explainable. | `report_service.py`; export tests | Partial; eligibility/grain/exclusion matrix is **BUSINESS DECISION REQUIRED**. | Yes | Unreliable evidence exports. |
| R-040 | PHASE1_PLATFORM | Audit platform | Technical actions have actor, computer, action, entity, before/after, result, error, version, and correlation evidence. | `application/audit.py`; audit finalization docs | Append-oriented JSONL implemented; completeness/retention remain open. | Yes | Platform actions cannot be reconstructed. |
| R-041 | PHASE1_PLATFORM | Audit assurance | Retention, access, export, completeness monitoring, and tamper-evidence standard must be defined for Phase 1 audit logs. | `audit.py`; `2026-08-26_AUDIT_EVENTS_FINALIZATION.md` | Partial — **BUSINESS DECISION REQUIRED**. | Yes | Evidence may fail compliance review. |
| R-042 | PHASE1_PLATFORM | Read-only safety | Phase 1 must technically prevent authoritative lifecycle mutation, downstream writes, automatic business decisions, and live provisioning. | `ARCHITECTURE.md`; `admin_routes.py`; `employee_routes.py` | **Approved technical acceptance requirement.** Existing writer-like routes create a repository contradiction — **CONTRADICTION**; add gate/composition boundary, do not refactor here. | No — acceptance is approved; Security must verify | Accidental authoritative change from an auditor platform. |
| R-043 | PHASE1_PLATFORM | Authentication/authorization | Access to sensitive analytical IAM data must use production-approved authentication and least-privilege authorization. | `identity.py`; `role_catalog.py`; `SECURITY.md` | Local login/CSV and dev auto-approval exist; production configuration is **BUSINESS DECISION REQUIRED**. | Yes | Unauthorized data exposure or mutation. |
| R-044 | PHASE1_PLATFORM | Roles | Requester/reviewer/auditor/admin roles may exist for platform access, but none may be treated as authoritative IAM decision authority in Phase 1. | `SECURITY.md`; `role_catalog.py`; `identity.py` | Role descriptions conflict — **CONTRADICTION**; Phase 1 matrix must be read/review only. | Yes | Privilege escalation or false approval. |

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|---|---|---|---|---|---|---|---|
| R-045 | PHASE1_PLATFORM | Analytical snapshot distribution | Technical publication/distribution of a read-only analytical database may be used for multi-user operation with checksum/concurrency controls. | `runtime_sync.py`; `admin_routes.py` | Backup, checksum, atomic sync, and publication signal exist. This is not business-change publication. | Yes | Users may receive inconsistent analytical versions. |
| R-046 | PHASE1_PLATFORM | Notifications | Operational notifications remain local outside the database; only analytical database snapshot publication may generate a global signal, throttled at five minutes. | `local_notification_store.py`; `shared_notification_store.py`; notification tests | Implemented; privacy/retention/backup of local files remain **BUSINESS DECISION REQUIRED**. | Yes | Cross-user noise, leakage, or lost operational evidence. |
| R-047 | PHASE1_PLATFORM | Recovery | Backup, restore, version, checksum, and atomic refresh behavior protect analytical evidence and runtime operation. | `runtime_sync.py`; recovery tests | Implemented/partially tested; RPO/RTO/retention are **BUSINESS DECISION REQUIRED**. | Yes | Wrong snapshot or incomplete evidence may be restored. |
| R-048 | PHASE2_WRITER | Request workflow | Request state machines, approvals, apply, publication queues, and ITSM-oriented workflows are writer/change-management functionality. | `change_request.py`; `request_service.py`; `shared_request_service.py` | Existing capabilities are retained as future documentation — **DEFERRED — PHASE2_WRITER**. | No for Phase 1 | Phase 1 sign-off becomes dependent on future workflow. |
| R-049 | FUTURE_INTEGRATION | Salesforce live integration | Salesforce API authentication, live create/update, and account provisioning are not Phase 1 capabilities. | `salesforce_export_service.py`; Salesforce guide; REQ-061 | No live API found — **DEFERRED — FUTURE_INTEGRATION**. | No for Phase 1 | Manual evidence export may be mistaken for provisioning. |
| R-050 | FUTURE_INTEGRATION | Other downstream integrations | AD/Azure, CXOne, NICE, SQL, orchestrator, and ITSM live connectors are future integration boundaries. | requirements REQ-060..066; README | Placeholders/future requirements; no Phase 1 live mutation. | No for Phase 1 | Hidden dependencies can block an otherwise valid audit release. |
| R-051 | FUTURE_INTEGRATION | Source platform writes | Modifying Fieldglass, AccessNow, Master Roster, Roster Lock, or other authoritative source platforms is outside Phase 1. | architecture clarification; README; source services | Source services ingest/read; any write path is future writer/integration scope. | No for Phase 1 | Auditor could mutate source truth. |
| R-052 | OUT_OF_SCOPE | Authoritative IAM decisions | Hire/reject hire, terminate, reactivate, access change, provisioning, and business approval are not current UAT requirements. | architecture clarification; old request/approval docs | Explicitly outside Phase 1; preserve future design separately. | No for Phase 1 | UAT may validate a decision that only humans/source systems may make. |

## Phase 1 finding classification model

For each matched field, population comparison, lifecycle observation, or source absence, the analytical platform should classify evidence—not choose the business outcome:

| Finding class | Meaning for UAT |
|---|---|
| CONSISTENT | Observations agree within the approved comparison rules. |
| EXPECTED_DIFFERENCE | Difference is known and valid for the population/process. |
| SOURCE_LATENCY | Difference may be explained by an approved refresh/arrival window. |
| INSUFFICIENT_EVIDENCE | Available data cannot safely support a conclusion. |
| DISCREPANCY | Evidence differs beyond the accepted rule/window and needs human review. |
| POLICY_RELEVANT_FINDING | Evidence may require a human business decision; Phase 1 does not make it. |